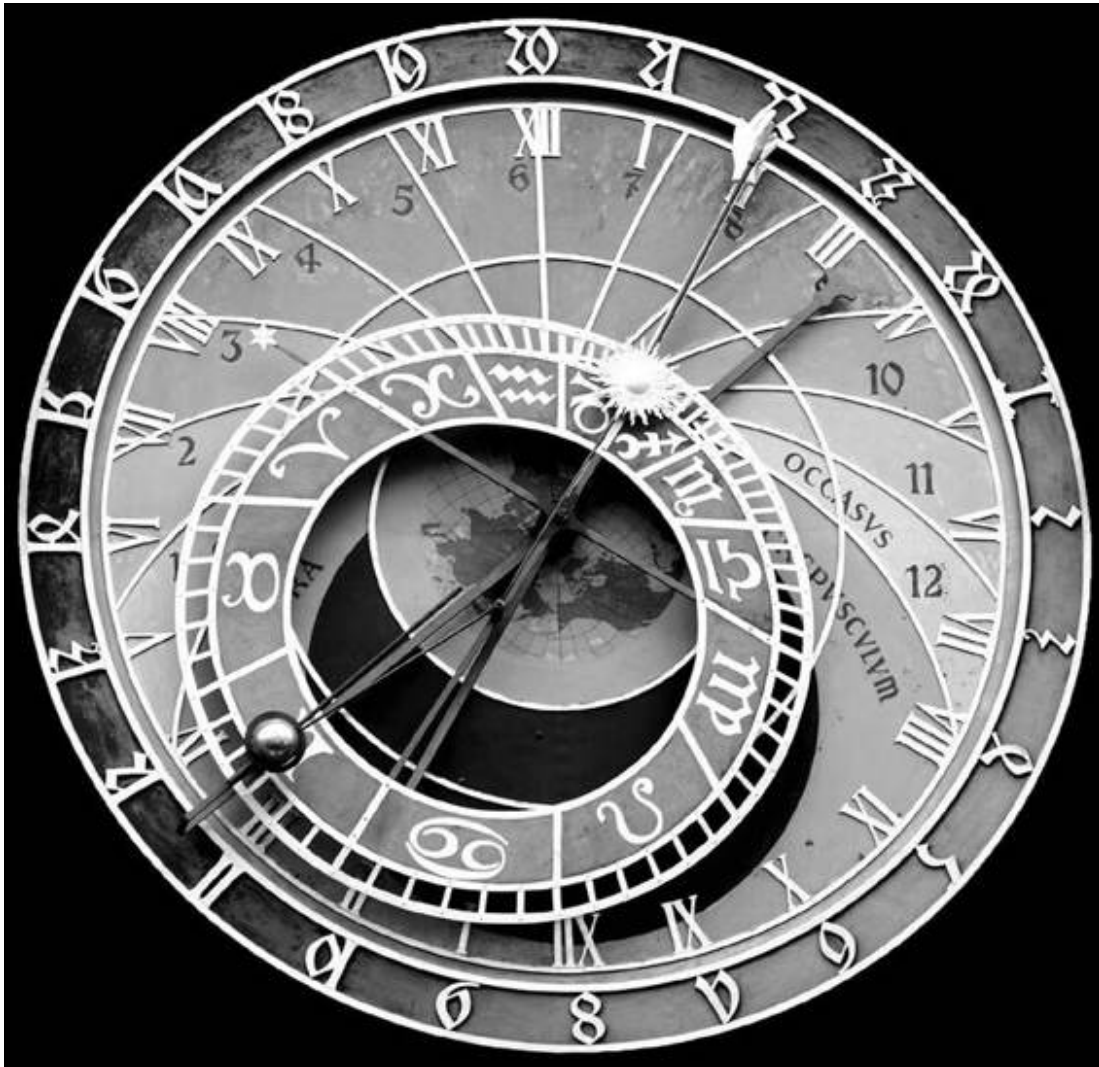


The problem of induction



Astronomical clocks such as this one in Prague show the positions of the heavenly bodies based on countless observations – but can we be sure their movements are predictable?

In casting doubt on the notion of causality and our certainty about our knowledge of the world (see [Hume and causality](#) and [Hume's fork](#)), Hume challenged the basis of modern scientific methodology. He simply pointed out that just because something always *has* happened, we cannot be sure it always *will* happen: we cannot say that something *causes* it to happen. To deny, for example, that the Sun will rise tomorrow doesn't produce a contradictory relation of ideas and isn't an impossible matter of fact – so we

have no rational grounds for believing it. This means that induction – deriving a general rule from a series of individual instances – is not logically valid, raising a huge problem for science, which uses induction to form theories and general laws based on empirical observations and repeatable experiments. This ‘problem of induction’ worried philosophers until Karl Popper proposed a solution in the 20th century (see [Modern logic](#)). Hume addressed it in characteristically good-humoured fashion, saying that we just have to deal in hopeful probabilities, not certainties, and should let custom be our guide.